TSM Taiwan Semiconductor Manufacturing Company Limited — 个股深度研究 (专业中文版) v20261016-0756

> 数据截至 2026-10-16 07:56 SGT｜价格 $422.25｜市值 2,190.0 bn USD｜财报货币 TWD｜非投资建议。事实=CONFIRMED/DATA；推断=INFERENCE；假设=ASSUMPTION/HYPOTHESIS。

1. 投资论点 Investment Thesis (HYPOTHESIS)

台积电是 AI 芯片无可替代的制造瓶颈，拥有定价权；主要风险是地缘政治与海外厂毛利稀释。财报以新台币计。

- **多头驱动**: N2 ramp + price increases; CoWoS doubling; AI accelerators >30% revenue｜N2 爬坡+涨价；CoWoS 倍增；AI 加速器占比超 30%
- **空头驱动**: Geopolitical shock, US tariffs on chips, Intel/Samsung regain share｜地缘冲击、美国芯片关税、英特尔/三星夺回份额
- **证伪条件 Falsifiers**: GM falls below long-term target band; monthly revenue y/y decelerates sharply; capex cut｜毛利率跌破长期目标区间；月营收同比骤降；Capex 下修

2. 产品、客户与收入模式

- 主题 Theme: Foundry / Advanced packaging 晶圆代工/先进封装
- 业务分部 Segments: HPC/AI 高性能计算；Smartphone 智能手机；IoT/Auto/DCE 其他
- 公司描述 (DATA, Yahoo/公司申报): Taiwan Semiconductor Manufacturing Company Limited, together with its subsidiaries, manufactures, packages, tests, and sells integrated circuits and other semiconductor devices in Taiwan, China, Europe, the Middle East, Africa, Japan, the United States, and internationally. It provides various wafer fabrication processes, such as processes to manufacture complementary metal- oxide-semiconductor (CMOS) logic, mixed-signal, radio frequency, embedded memory, bipolar CMOS mixed-signal, and others. The company also involved in providing customer and engineering support services; manufacturing of masks; investment in technology start-up companies; research, designing, developing, manufacturing, packaging, testing, and sale of color filters; and investment activities. Its products are used in high performance computing, smartphones, Internet of things, automotive, and digital consumer electroni

- 主要客户: NVDA, Apple, AMD, Broadcom, Qualcomm, MediaTek, Marvell, hyperscaler ASICs｜NVDA、苹果、AMD、博通、高通、联发科、Marvell、云厂 ASIC

3. 行业、竞争、监管与供应链 (ANALYST FRAMEWORK)

- 供应商/上游: ASML (EUV), AMAT/LRCX/TEL/KLAC, Shin-Etsu/SUMCO (wafers), chemicals｜ASML、设备商、硅片、化学品
- 竞争者 (含业外): Samsung Foundry, Intel Foundry (18A/14A), SMIC｜三星代工、英特尔代工、中芯国际
- 瓶颈/监管: CoWoS/SoIC capacity, N2/A16 ramp, overseas fab cost (Arizona/Japan), Taiwan geopolitics, power/water｜CoWoS/SoIC 产能、N2/A16 爬坡、海外厂成本、台海地缘、电力水资源
- 关键 KPI: Monthly revenue, GM%, HPC mix, N3/N2 mix, capex, CoWoS capacity｜月营收、毛利率、HPC 占比、先进制程占比、Capex、CoWoS 产能
- 零部件现货/合约价: 本版未接入付费价格源(TrendForce/DRAMeXchange 等)——标注 **UNAVAILABLE**，在 KPI 监控中以定性方式跟踪。

4. 历史财务与现金流 (CONFIRMED, 季度, TWD)

季度	收入	毛利	营业利润	净利润	FCF	营业利润率
2026-06-30	1,270.4 bn TWD	860.3 bn TWD	766.6 bn TWD	706.6 bn TWD	275.0 bn TWD	60.3%
2026-03-31	1,134.1 bn TWD	751.3 bn TWD	658.9 bn TWD	572.5 bn TWD	347.3 bn TWD	58.1%
2025-12-31	1,046.1 bn TWD	652.0 bn TWD	564.9 bn TWD	485.5 bn TWD	364.0 bn TWD	54.0%
2025-09-30	989.9 bn TWD	588.5 bn TWD	500.7 bn TWD	452.3 bn TWD	138.4 bn TWD	50.6%
2025-06-30	933.8 bn TWD	547.4 bn TWD	463.4 bn TWD	398.3 bn TWD	199.8 bn TWD	49.6%
2025-03-31	N/A	N/A	N/A	N/A	N/A	N/A

资产负债: 现金 3,518.0 bn TWD；总负债 1,068.6 bn TWD；净现金 2,449.5 bn TWD。TTM FCF 1,124.7 bn TWD（FCF 率 25.3%）。

5. 最新财报: 实际 vs 一致预期 vs 指引

季度	EPS 实际	EPS 一致预期	超预期%
2025-12-31	3.14	2.97556	0.0553
2026-03-31	3.49	3.33393	0.0468
2026-06-30	4.31	3.88658	0.1089
2026-09-30	4.6	4.52449	0.0167

- 管理层指引 Guidance: 自动数据源不含指引——**UNAVAILABLE（待从公司 IR 新闻稿录入）**。
- 下次财报日 (Yahoo, TENTATIVE 直至公司确认): **2026-10-15**；本季 EPS 一致预期 5.07666（8 位分析师）；收入一致预期 1,700.4 bn TWD。

6. 未来 2/4/6/8 季度驱动与价格合理性

期限	情景	期限价格现值	相对现价
2Q	Bear	$132.89	-68.5%
2Q	Base	$277.41	-34.3%
2Q	Bull	$448.79	6.3%
4Q	Bear	$134.19	-68.2%
4Q	Base	$308.58	-26.9%
4Q	Bull	$528.12	25.1%
6Q	Bear	$135.58	-67.9%
6Q	Base	$343.56	-18.6%
6Q	Bull	$621.62	47.2%
8Q	Bear	$137.03	-67.5%
8Q	Base	$382.77	-9.3%
8Q	Bull	$731.68	73.3%

现价隐含的 EV/NTM-FCF (Base 路径):	期限
2Q	36.6x
4Q	31.0x
6Q	26.2x
8Q	22.1x

解读: 如果 8 季度后 Base 情景所需倍数仍远高于 25x，说明价格依赖更远期的增长。

7. 三情景 Bear / Base / Bull (HYPOTHESIS)

情景	Y1 增速	长期 FCF 率	WACC	永续 g	概率	每股价值	相对现价
Bear	17.6%	24.0%	12.4%	2.5%	25.0%	$132.47	-68.6%
Base	35.2%	30.0%	11.4%	3.0%	50.0%	$341.27	-19.2%
Bull	45.8%	34.5%	10.9%	3.5%	25.0%	$634.70	50.3%

概率加权价值 **$362.43**（相对现价 -14.2%）。

8. 估值: DCF / Reverse DCF / ROIC-WACC / 同业 / 敏感度

- DCF Base: EV 54,784.3 bn TWD；终值占比 56.2%。
- **Reverse DCF 反向 DCF**: 在 Base 利润率与 WACC 下，现价隐含 Y1 收入增速 ≈ **41.4%**；一致预期 FY+1 增速 35.2%。
- WACC (CAPM) = rf 5.2% + β 1.24 × ERP 5.0% = 11.4%。
- SOTP segments 分部: HPC/AI 高性能计算; Smartphone 智能手机; IoT/Auto/DCE 其他. **Status: framework only** — segment revenues are INPUT REQUIRED from the latest 10-K/10-Q segment note (not auto-fed, left blank to avoid fabrication)｜分部收入需按最新财报手动录入，未自动填充以避免编造。
- 同业比较 Comps (DATA): | Ticker | 市值 | EV/Sales | Fwd P/E | 收入增速 | 毛利率 | FCF 收益率 | |—|—|—|—|—|—|—| | TSM | 2,190.0 bn USD | 3.5x | 18.4x | 36.0% | 64.2% | 1.6% | | INTC | 504.1 bn USD | 9.1x | 45.3x | 25.4% | 38.9% | 1.0% | | 005930.KS | 1,367,503.8 bn USD | 2.5x | 2.9x | 130.0% | 57.5% | 5.1% | | UMC | 52.5 bn USD | -0.1x | 23.8x | 17.0% | 30.6% | 70.3% | | GFS | 27.2 bn USD | 3.8x | 18.5x | 5.8% | 27.1% | 2.8% |
- 敏感度: 见 Excel `Sensitivity` 表 (WACC × 永续增长率)。

9. 风险、反方论证与证伪

- 反方观点 Counter-thesis: Geopolitical shock, US tariffs on chips, Intel/Samsung regain share｜地缘冲击、美国芯片关税、英特尔/三星夺回份额
- 证伪指标: GM falls below long-term target band; monthly revenue y/y decelerates sharply; capex cut｜毛利率跌破长期目标区间；月营收同比骤降；Capex 下修

- 模型风险: 长期利润率与 WACC 为假设；终值占比高时估值对假设极敏感。

10. Source Ledger 来源账本

Item	Value	Source/time
Price 股价	422.2	Yahoo Finance quote, fetched 2026-10-16T07:56 SGT
Market cap 市值	2.19e+12	Yahoo info.marketCap 2026-10-16T07:56
Cash 现金	3.518e+12	Yahoo info.totalCash (latest filing) 2026-10-16T07:56
Debt 负债	1.069e+12	Yahoo info.totalDebt (latest filing) 2026-10-16T07:56
Revenue TTM 收入	4.44e+12	Yahoo info.totalRevenue (TTM, TWD) 2026-10-16T07:56
FCF TTM 自由现金流	1.125e+12	Sum of last 4 quarterly FCF (q_cash); quarters ['2026-06-30', '2026-03-31', '2025-12-31', '2025-09-30'] (TWD)
Beta	1.239	Yahoo info.beta (5Y monthly)
Risk-free 10Y 无风险利率	0.0519	FRED DGS10 2026-10-14
FX USD per TWD	0.03093	Yahoo FX quote
Consensus rev growth FY0	0.4493	Yahoo revenue_estimate 0y
Consensus rev growth FY+1	0.3521	Yahoo revenue_estimate +1y

11. Assumption Register 假设登记表

Assumption	Value	Rationale
ERP 股权风险溢价	0.05	ASSUMPTION: 5.0% house mature-market ERP
Beta clamp	[0.8,1.6] → 1.24	ASSUMPTION: limit extreme betas

Assumption	Value	Rationale
Tax rate (ROIC)	0.18	ASSUMPTION: blended cash tax 18%
Year-1 growth (Base)	0.3521	Consensus FY+1 revenue growth (Yahoo revenue_estimate '+1y'); capped at 100%
Growth fade	linear Y1→terminal by Y10	ASSUMPTION
Starting FCF margin	0.2533	DATA: FCF TTM / Revenue TTM, floored -50%
Long-run FCF margin (Base)	0.3	ASSUMPTION (analyst framework, reached by Y5; see profile)
Bear/Bull Y1 growth multiplier	0.5x / 1.3x	ASSUMPTION
Bear/Bull margin multiplier	0.8x / 1.15x	ASSUMPTION
WACC adj Bear/Bull	+1.0pp / -0.5pp	ASSUMPTION
Terminal g Bear/Base/Bull	2.5% / 3.0% / 3.5%	ASSUMPTION
Scenario probabilities	25% / 50% / 25%	ASSUMPTION
Horizon exit EV/NTM-FCF Bear/Base/Bull	15x / 25x / 35x	ASSUMPTION
Per-share conversion	value × fx ÷ market cap × price	METHOD: avoids ADR/share-class mismatch

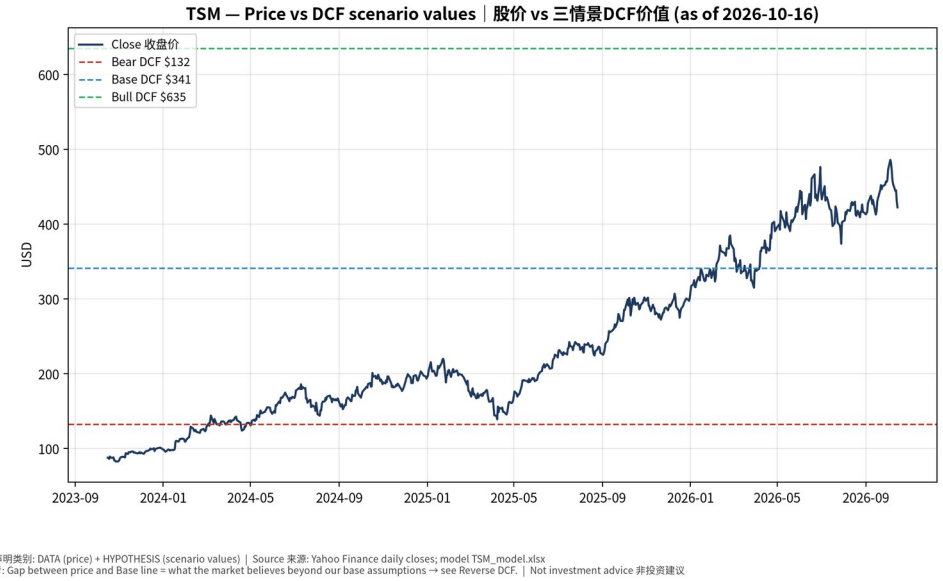
12. Model Provenance 模型溯源

- Model file: `TSM_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions)
- Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx)
- Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT
- Version: v20261016-0756. Prior versions are never overwritten (see company history page).

13. 版本与事件时间轴

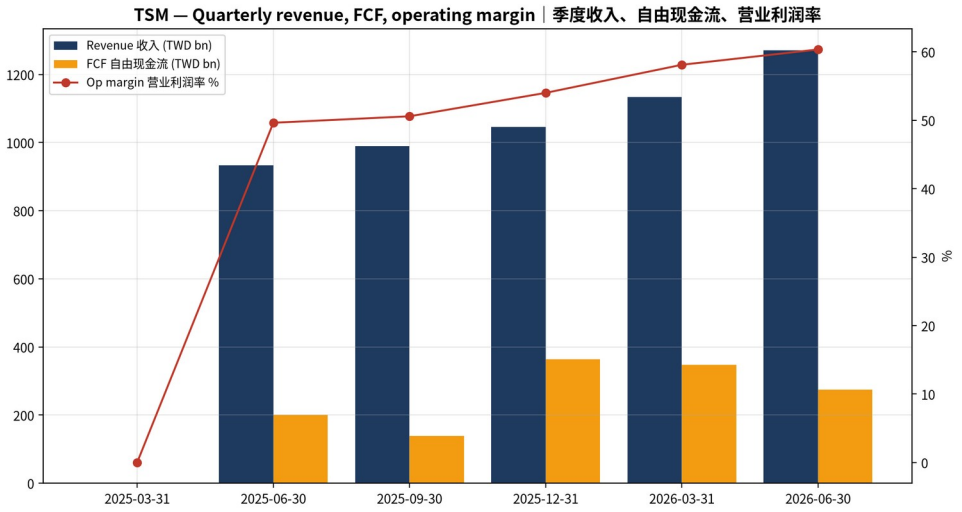
- v20261016-0756: 首版生产系统报告 (27 家同步)。下一事件: 财报 2026-10-15 (TENTATIVE)。

图片报告 Image Report



1_price_vs_value

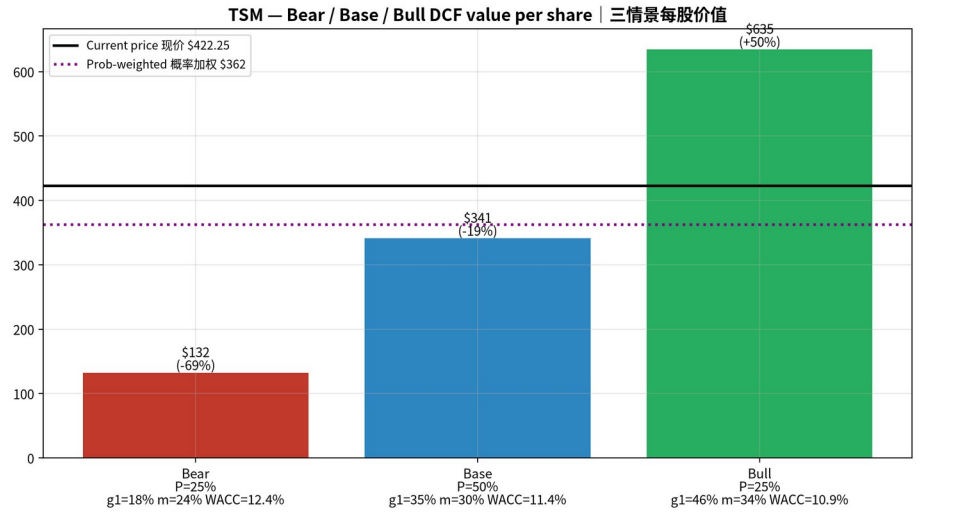
股价vs 三情景价值Price vs scenario values



Claim class 声明类别: CONFIRMED (reported) | Source 来源: Company filings via Yahoo Finance quarterly statements
Thinking 思考: Is FCF growing with revenue? Divergence = capex/working-capital intensity → check cash-flow statement. | Not investment advice 非投资建议

2_financials

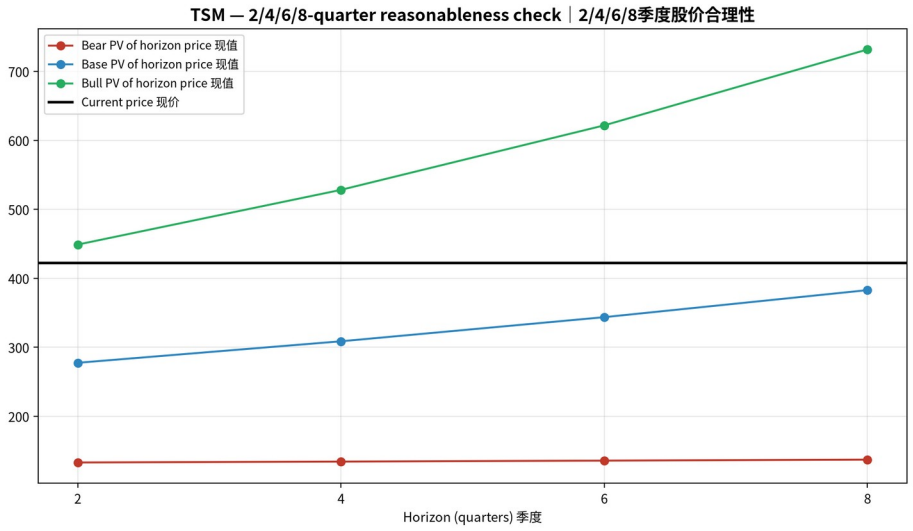
季度财务 Quarterly financials



Claim class 声明类别: HYPOTHESIS (model output) | Source 来源: Model (live formulas) — inputs in Sources/Assumptions sheets
Thinking 思考: Which assumption moves value most? Check Sensitivity sheet; scenario width = uncertainty. | Not investment advice 非投资建议

3_scenarios

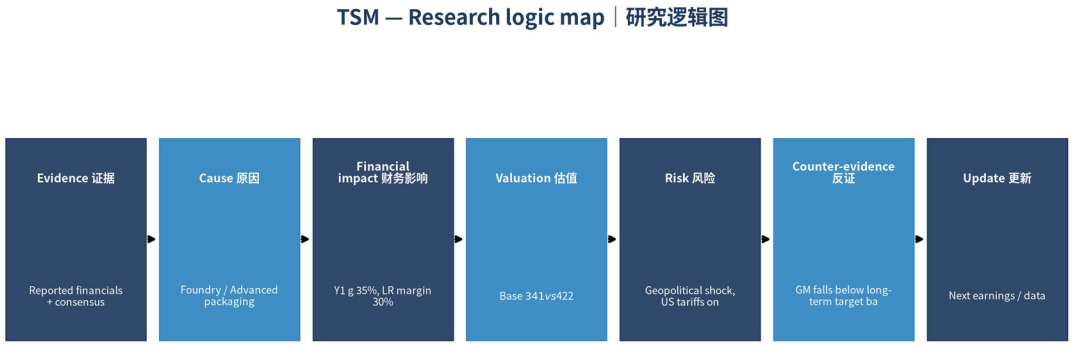
三情景 Scenarios



Claim class 声明类别: HYPOTHESIS | Source 来源: Model Horizons sheet; exit EV/NTM-FCF 15x/25x/35x (ASSUMPTION)
Thinking 思考: If even Bull PV < price at 8Q, price requires assumptions beyond Bull within 2 years. | Not investment advice 非投资建议

4_horizons

2/4/6/8 季度检验Horizon check



Reverse DCF 反向DCF: price implies Y1 growth ≈ 41%

Consensus FY+1 growth 一致预期: 35% → gap = expectation the price embeds 价格隐含预期差

Claim class 声明类别: INFERENCE / HYPOTHESIS | Source 来源: Profile (analyst framework) + model
Thinking 思考: Read left→right: each box must have evidence; the falsifier box tells you what would prove the thesis wrong. | Not investment advice 非投资建议

5_logic_map

研究逻辑图Logic map

TSM Taiwan Semiconductor Manufacturing Company Limited — Equity Deep Dive (Professional English) v20261016-0756

Data as of 2026-10-16 07:56 SGT | Price $422.25 | Mkt cap 2,190.0 bn USD | Reporting currency TWD | Not investment advice.

1. Investment thesis (HYPOTHESIS)

TSMC is the irreplaceable manufacturing bottleneck for AI chips with pricing power; key risks are geopolitics and overseas-fab margin dilution. Financials are reported in TWD.

- **Bull drivers**: N2 ramp + price increases; CoWoS doubling; AI accelerators >30% revenue｜N2 爬坡+涨价；CoWoS 倍增；AI 加速器占比超 30%
- **Bear drivers**: Geopolitical shock, US tariffs on chips, Intel/Samsung regain share｜地缘冲击、美国芯片关税、英特尔/三星夺回份额
- **Falsifiers**: GM falls below long-term target band; monthly revenue y/y decelerates sharply; capex cut｜毛利率跌破长期目标区间；月营收同比骤降；Capex 下修

2. Products, customers, revenue model

- Theme: Foundry / Advanced packaging 晶圆代工/先进封装 | Segments: HPC/AI 高性能计算; Smartphone 智能手机; IoT/Auto/DCE 其他
- Business description (DATA): Taiwan Semiconductor Manufacturing Company Limited, together with its subsidiaries, manufactures, packages, tests, and sells integrated circuits and other semiconductor devices in Taiwan, China, Europe, the Middle East, Africa, Japan, the United States, and internationally. It provides various wafer fabrication processes, such as processes to manufacture complementary metal- oxide-semiconductor (CMOS) logic, mixed-signal, radio frequency, embedded memory, bipolar CMOS mixed-signal, and others. The company also involved in providing customer and engineering support services; manufacturing of masks; investment in technology start-up companies; research, designing, developing, manufacturing, packaging, testing, and sale of color filters; and investment activities. Its products are used in high performance computing, smartphones, Internet of things, automotive, and digital consumer electroni
- Customers: NVDA, Apple, AMD, Broadcom, Qualcomm, MediaTek, Marvell, hyperscaler ASICs｜NVDA、苹果、AMD、博通、高通、联发科、Marvell、云厂 ASIC

3. Industry, competition, regulation, supply chain (ANALYST FRAMEWORK)

- Suppliers: ASML (EUV), AMAT/LRCX/TEL/KLAC, Shin-Etsu/SUMCO (wafers), chemicals｜ASML、设备商、硅片、化学品
- Competitors: Samsung Foundry, Intel Foundry (18A/14A), SMIC｜三星代工、英特尔代工、中芯国际
- Bottlenecks/regulation: CoWoS/SoIC capacity, N2/A16 ramp, overseas fab cost (Arizona/Japan), Taiwan geopolitics, power/water｜CoWoS/SoIC 产能、N2/A16 爬坡、海外厂成本、台海地缘、电力水资源
- KPIs: Monthly revenue, GM%, HPC mix, N3/N2 mix, capex, CoWoS capacity｜月营收、毛利率、HPC 占比、先进制程占比、Capex、CoWoS 产能
- Component spot/contract prices: paid feeds not connected — **UNAVAILABLE**, tracked qualitatively.

4. Historical financials & cash flow (CONFIRMED, quarterly, TWD)

Quarter	Revenue	Gross profit	Op income	Net income	FCF	Op margin
2026-06-30	1,270.4 bn TWD	860.3 bn TWD	766.6 bn TWD	706.6 bn TWD	275.0 bn TWD	60.3%
2026-03-31	1,134.1 bn TWD	751.3 bn TWD	658.9 bn TWD	572.5 bn TWD	347.3 bn TWD	58.1%
2025-12-31	1,046.1 bn TWD	652.0 bn TWD	564.9 bn TWD	485.5 bn TWD	364.0 bn TWD	54.0%
2025-09-30	989.9 bn TWD	588.5 bn TWD	500.7 bn TWD	452.3 bn TWD	138.4 bn TWD	50.6%
2025-06-30	933.8 bn TWD	547.4 bn TWD	463.4 bn TWD	398.3 bn TWD	199.8 bn TWD	49.6%
2025-03-31	N/A	N/A	N/A	N/A	N/A	N/A

Cash 3,518.0 bn TWD; debt 1,068.6 bn TWD; TTM FCF 1,124.7 bn TWD (FCF margin 25.3%).

5. Latest earnings: actual vs consensus vs guidance

Quarter	EPS actual	EPS consensus	Surprise
2025-12-31	3.14	2.97556	0.0553
2026-03-31	3.49	3.33393	0.0468

Quarter	EPS actual	EPS consensus	Surprise
2026-06-30	4.31	3.88658	0.1089
2026-09-30	4.6	4.52449	0.0167

- Management guidance: not in automated feed — **UNAVAILABLE (to be entered from IR release)**.
- Next earnings (Yahoo; TENTATIVE until company-confirmed): **2026-10-15**; current-quarter EPS consensus 5.07666 (8 analysts); revenue consensus 1,700.4 bn TWD.

6. 2/4/6/8-quarter drivers & price reasonableness

Horizon	Scenario	PV of horizon price	vs current
2Q	Bear	$132.89	-68.5%
2Q	Base	$277.41	-34.3%
2Q	Bull	$448.79	6.3%
4Q	Bear	$134.19	-68.2%
4Q	Base	$308.58	-26.9%
4Q	Bull	$528.12	25.1%
6Q	Bear	$135.58	-67.9%
6Q	Base	$343.56	-18.6%
6Q	Bull	$621.62	47.2%
8Q	Bear	$137.03	-67.5%
8Q	Base	$382.77	-9.3%
8Q	Bull	$731.68	73.3%

Current-price implied EV/NTM-FCF on Base path:	Horizon
2Q	36.6x
4Q	31.0x
6Q	26.2x
8Q	22.1x

7. Bear / Base / Bull (HYPOTHESIS)

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/ share	vs price
Bear	17.6%	24.0%	12.4%	2.5%	25.0%	$132.47	-68.6%
Base	35.2%	30.0%	11.4%	3.0%	50.0%	$341.27	-19.2%

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/ share	vs price
Bull	45.8%	34.5%	10.9%	3.5%	25.0%	$634.70	50.3%

Probability-weighted value **$362.43** (-14.2% vs price).

8. Valuation: DCF / Reverse DCF / ROIC-WACC / comps / sensitivity

- Base DCF EV 54,784.3 bn TWD; terminal share 56.2%.
- **Reverse DCF**: at Base margin/WACC the price implies Y1 revenue growth ≈ **41.4%** vs consensus FY+1 35.2%.
- WACC (CAPM) = rf 5.2% + β 1.24 × ERP 5.0% = 11.4%.
- SOTP segments: HPC/AI 高性能计算; Smartphone 智能手机; IoT/Auto/DCE 其他 — framework only; segment revenue inputs pending from filings.
- Comps (DATA): see table in Chinese section / Excel `Comps`.
- Sensitivity: Excel `Sensitivity` (WACC × terminal g).

9. Risks, counterarguments, falsifiers

Geopolitical shock, US tariffs on chips, Intel/Samsung regain share｜地缘冲击、美国芯片关税、英特尔/三星夺回份额 | Falsifiers: GM falls below long-term target band; monthly revenue y/y decelerates sharply; capex cut｜毛利率跌破长期目标区间；月营收同比骤降；Capex 下修

10–12. Source Ledger / Assumption Register / Model Provenance

See tables above (shared, bilingual). - Model file: `TSM_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions) - Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx) - Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT - Version: v20261016-0756. Prior versions are never overwritten (see company history page).